



**POLITECNICO
MILANO 1863**

**COMPUTER SCIENCE AND ENGINEERING
INTERNET OF THINGS (IoT)
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Homework IEEE 802.15.4

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Exercise 1

Assume that the number of vibration peaks detected in each 10-second window follows a Poisson distribution with average rate: $\lambda = 0.4 \text{ peaks/window}$.

The system is operated with IEEE 802.15.4 in beacon-enable mode, CFP only.

Assume: $R = 250 \text{ kbps}$, $L = 128 \text{ bytes}$, where R is the nominal bit rate and L is the packet size.

Assume that:

- one packet fits exactly in one CFP slot;
- the CFP is dimensioned according to the expected traffic load;
- the reporting period is 10 seconds;
- the monitoring system is composed of 1 PAN coordinator and 3 vibration monitoring nodes.

```
// Global timer handle
declare timer_handle as esp_timer_handle_t

function setup_accelerometer():
    initialize_accelerometer(sampling_rate = 100 Hz)

function setup_timer():
    create_periodic_timer(callback = process_window,
                          period = 10 seconds)

function process_window(arg):
    vibration_peaks = count_vibration_peaks(last_10_seconds)

    if vibration_peaks == 0:
        payload = create_message(size = 512 bytes)
    else if vibration_peaks == 1:
        payload = create_message(size = 2 KB)
    else:
        payload = create_message(size = 4 KB)

    send_payload(payload)

function app_main():
    setup_accelerometer()
    setup_timer()

    loop forever:
        delay(100 ms)
```

Q1)

Compute the Probability Mass Function of the output rate: $P(r = r_0)$, $P(r = r_1)$, $P(r = r_2)$ where r_0 , r_1 , and r_2 are the output rates when the node detects 0, 1, or more than 1 vibration peaks, respectively.

1.1 Theory

For a Poisson random variable:

$$P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

where $\lambda = 0.4$.

1.2 PMF (Probability Mass Function)

$$P(r = r_0) = P(X = 0) = \frac{0.4^0 \cdot e^{-0.4}}{0!} = e^{-0.4} = 0.6703$$

$$P(r = r_1) = P(X = 1) = \frac{0.4^1 \cdot e^{-0.4}}{1!} = 0.4 \cdot e^{-0.4} = 0.2681$$

$$P(r = r_2) = P(X > 1) = 1 - P(X = 0) - P(X = 1) = 1 - 0.6703 - 0.2681 = 0.0616$$

Q2)

Based on the output rate PMF, compute a consistent CFP slot assignment for the system.

2.1 Output Rate

The reporting period of the monitoring application is fixed to 10 seconds, as specified in the pseudocode.

Therefore, each node generates one payload every 10 seconds.

For this reason, the output rate is computed by dividing the payload size by the reporting interval.

The output rates are calculated using the formula:

$$r_n = \frac{L_n [\text{bytes}] \times 8 [\text{bits/byte}]}{BI [\text{s}]} [\text{bits/s}]$$

where L_n is the payload size associated with the detected vibration condition and BI represents the reporting interval of the application.

Vibration Peaks	Message Size	Output Rate
0	512 [bytes]	$r_0 = \frac{512 [\text{bytes}] \times 8 [\text{bits/byte}]}{10 \text{ s}} = 409,6 [\text{bits/s}]$
1	2 KB = 2048 [bytes]	$r_1 = \frac{2048 [\text{bytes}] \times 8 [\text{bits/byte}]}{10 \text{ s}} = 1638,4 [\text{bits/s}]$
>1	4 KB = 4096 [bytes]	$r_2 = \frac{4096 [\text{bytes}] \times 8 [\text{bits/byte}]}{10 \text{ s}} = 3276,8 [\text{bits/s}]$

2.2 CFP Slot

The worst-case rate is $r_2 = 3276,8 [\text{bit/s}]$.

The minimum rate is $r_0 = 409,6 [\text{bit/s}]$.

Therefore:

$$N_{CFP, node} = \frac{\text{worst-case rate}}{\text{min rate}} = \frac{3276,8 [\text{bit/s}]}{409,6 [\text{bit/s}]} = 8 \text{ slots}$$

Q3)

Compute:

- T_s
- N slots in the CFP
- T_{active}
- $T_{inactive}$
- Duty Cycle

Equivalent Beacon Interval for CFP Slot Dimensioning

$$BI = \frac{L}{r_0} = \frac{128 \text{ [bytes]} \times 8 \text{ [bit/byte]}}{409,6 \text{ [bit/s]}} = 2,5 \text{ [s]}$$

Slot Time

The slot time is determined by the time required to transmit one packet:

$$T_s = \frac{L_s}{R} = \frac{128 \text{ [bytes]} \times 8 \text{ [bits/byte]}}{250 \text{ [kbps]}} = \frac{1024 \text{ [bits]}}{250.000 \text{ [bps]}} = 4,096 \text{ [ms]}$$

N Slots In The CFP

The number of required CFP slots for one node is:

$$N_{CFP,node} = 8 \text{ slots}$$

Since we have 3 vibration monitoring nodes:

$$N_{CFP,tot} = 8 \times 3 = 24 \text{ slots}$$

Active Time

The active time includes the first beacon signal, the CAP and the CFP slots.

Since the CAP here is equal to 0, the active time is:

$$T_{active} = (1 + 24) \times T_s = 25 \times 4,096 \text{ [ms]} = 102,4 \text{ [ms]}$$

Inactive Time

$$T_{inactive} = BI - T_{active} = 2,5 \text{ [s]} - 102,4 \text{ [ms]} = 2397,6 \text{ [ms]}$$

Duty Cycle

$$\eta = \frac{T_{active}}{BI} = \frac{102,4 \text{ [ms]}}{2,5 \text{ [s]}} = 0,04096 \approx 4,1 \%$$

Q4)

How many additional vibration monitoring nodes can be added while keeping the duty cycle below 10%?

To maintain a duty cycle below 10%, we should have:

$$\eta = \frac{T_{active}}{BI} < 0.1$$

Substituting the expressions:

$$\frac{(1 + N_{CFP}) \times T_s}{BI} < 0.1$$

Solving for N_{CFP} :

$$N_{CFP} < \frac{0.1 \times BI}{T_s} - 1 = \frac{0.1 \times 2,5 [s]}{4,096 [ms]} - 1$$
$$N_{CFP} < 60,035$$

Therefore, the maximum $N_{CFP} = 60 \text{ slots}$.

The number of vibration monitoring nodes that can be supported:

$$N_{nodes} = \frac{N_{CFP}}{N_{CFP,node}} = \frac{60}{8} = 7,5 \sim 7$$

Since the current system has 3 nodes, 4 additional vibration monitoring nodes can be added while maintaining the duty cycle below 10%.